Niseem Bhattacharya

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Portfolio | GitHub | LinkedIn

EDUCATION

Greenhill School, Addison, TX — Expected Graduation: May 2027
Senior; GPA: 4.01 Unweighted

Selected Coursework: AP Physics C, AP Statistics, Post-AP Multivariable Calculus, AP Calculus BC, AP Physics 2, AP Computer Science A, Honors Applied Computer Science, Honors UX Design, Advanced Topics Tutorial in Design & Innovation

Standardized Testing:
SAT: **1520 Superscore** (730 Reading & Writing, 790 Math)
AP Scores: AP Calculus BC **(5)**, AP Physics 2 **(5)**, AP Computer Science A **(5)**, AP United States Government and Politics **(5)**, AP Physics C: Mechanics (will take), AP Physics C: Electricity and Magnetism (will take), AP Statistics (will take)

Independent Academic Programs: Harvard Computer Society AI Bootcamp; Harvard CS50P; SMU Lyle Engineering Camp

HONORS & RECOGNITION

- **National Merit Commended Student** – recognized through the National Merit Scholarship Program.
- **Kathryn and Ashley H. Priddy Young Scientist Award** – Dallas Regional Science Fair; supported research participation in UT Southwestern's STARS program.
- **Certificate of Achievement – Outstanding Accomplishment in AP Physics 2** – Greenhill School.
- **AP Scholar with Honor** – College Board recognition.
- **High Honor Roll** – Greenhill School.

RESEARCH & PROFESSIONAL EXPERIENCE

UT Southwestern Medical Center – STARS Program, Dallas, TX — June 2026 – July 2026
AI/ML Research Intern, Pool Lab – Dr. Allan-Hermann Pool — *40 hrs/wk, 8 wks/yr*

- Selected for UT Southwestern's competitive eight-week STARS research program.
- Combined human and mouse single-cell RNA sequencing data to study neuronal populations involved in transmitting pain signals from the spinal cord to the brain.
- Designed protein binders for KCNK13 using BindCraft, ultimately generating 13 accepted designs across four protein backbones and evaluating structural placement around an extracellular binding region.
- Built a separate GPR139 computational design workflow using MetaGen, RIFDock, ProteinMPNN, and AlphaFold, producing 142 promising docking candidates for further evaluation.

East Texas A&M University / Greenhill Advanced Topics Tutorial — Summer 2026 – Present
Signal Processing Researcher – Dr. Gerald Fudge — *12 hrs/wk, 8 wks/yr*

- Built Python simulations of frequency-varying chirp signals embedded in complex Gaussian noise and generated time-frequency spectrograms across progressively lower signal-to-noise ratios.
- Developed binary detection masks from spectrogram data and reduced SNR until signals fragmented, creating a controlled testbed for evaluating detection and clustering methods.
- Tested K-means, spectral, and agglomerative clustering on fragmented time-frequency detections; advancing toward wideband sensing and Nyquist Folding Receiver (NyFR) hardware experimentation.

Southern Methodist University, Dallas, TX — May 2025 – Present
Machine Learning Researcher & App Developer – Prof. Kasi Periyasamy — *10 hrs/wk, 40 wks/yr*

- Developed **DiaTrend**, a full-stack diabetes risk prediction and monitoring application, expanding initial machine-learning research into a tool for tracking patient health over time.
- Cleaned and modeled 986 patient records using logistic regression, scaling, imputation, and stratified validation; achieved ROC-AUC of approximately 0.85 while restructuring the pipeline to reduce data leakage.
- Developed application features for risk estimation, glucose logging, longitudinal visualization, progression tracking, and

monitoring so that model output could be interpreted over time rather than viewed as a one-time score.
- Presented the research through the Dallas Regional Science Fair and received the Kathryn and Ashley H. Priddy Young Scientist Award.

Datacolor.ai & InzpireU Summer 2025
AI & Data Engineering Intern *30 hrs/wk, 8 wks/yr*

- Contributed to **GrantMate**, an AI-assisted grant platform used by five nonprofit organizations and supporting approximately 56 grant applications.
- Built automated web-scraping and data-ingestion pipelines that streamlined grant research and were estimated to save more than 200 hours of manual work.
- Developed grant-search and document-analysis features using Flask, React, MySQL, APIs, and the OpenAI API, including compatibility scoring and automated PDF information extraction.

SELECTED PROJECTS & INDEPENDENT WORK

MedSight – HackSMU VII | **Devpost** April 2026 | 24 hrs/wk, 1 wk/yr

- Built MedSight with a team at HackSMU VII, using smart-glasses input to passively capture patient behavior and convert observations into structured information for physicians.
- Developed a computer-vision and language-model pipeline to extract behavioral signals and organize them into clinically useful summaries and trends.
- Integrated real-time voice functionality through ElevenLabs; won **Best Use of ElevenLabs** at HackSMU VII.

Python Chess Puzzle Auto-Solver | **GitHub** | **Demo** November 2025 | 10 hrs/wk, 2 wks/yr

- Built an autonomous chess-puzzle solver for Harvard CS50P using OpenCV, Stockfish, and PyAutoGUI to connect computer vision, chess analysis, and desktop automation.
- Detected on-screen board positions, reconstructed them in FEN notation, generated engine-selected moves with Stockfish, and executed them automatically on screen.

Personal Portfolio Website | **Live Site** | **GitHub** 2026 | 5 hrs/wk, 8 wks/yr

- Designed and coded an interactive portfolio using **React, JavaScript, HTML, CSS, and Vite** to showcase research, projects, and academic interests.

LEADERSHIP & COMMUNITY INVOLVEMENT

Table of Hope Dallas January 2025 – Present
Co-Founder & Co-President *5 hrs/wk, 40 wks/yr*

- Co-founded and co-lead a student-run service organization addressing food insecurity in Dallas and India.
- Helped scale the organization to serve **10,000+ people** and raise **$6,000+** for food distribution and community outreach.
- Built the organization's website, coordinated volunteers and distributions, and used recipient feedback to guide future outreach.

HornetHacks 2026 – Present
Co-Founder *10 hrs/wk, 16 wks/yr*

- Co-founded an **MLH-approved** high-school hackathon for students across the Dallas–Fort Worth area.
- Secured sponsor support from **Google Gemini and DigitalOcean** and built the event website and registration platform.
- Currently leading sponsor coordination, participant outreach, school logistics, and event operations ahead of the hackathon.

Young Men's Service League (YMSL) *Grades 9–11*
Member *2 hrs/wk, 10 wks/yr*

- Completed **50+ hours of community service** through YMSL, participating in volunteer projects supporting local nonprofit organizations and community needs.

SCHOOL INVOLVEMENT & ACADEMIC SERVICE

Greenhill Math Lab
Volunteer Peer Tutor – Junior Year | 1 hr/wk, 25 wks/yr

- Volunteered in Greenhill's Math Lab during junior year, helping other students work through homework, course concepts, and challenging math problems.

Philosophy Club
Member – Junior Year | 1 hr/wk, 20 wks/yr

- Participated during junior year, discussing philosophical questions and ideas outside my regular coursework.

Chess Club
Student Leader – Junior Year | 2 hrs/wk, 20 wks/yr

- Took an active role in Greenhill's Chess Club during junior year and continued playing chess independently outside of school.

CREATIVE, MUSIC & ATHLETICS

Independent Sports Video Editing – @editsbyseem
Video Editor *10 hrs/wk, 20 wks/yr | 3 years*

- Self-taught **Adobe Premiere Pro and After Effects** to create basketball and football highlight edits, motion graphics, and short-form sports content.
- Produced videos for multiple **Division I athletes and an NBA player**, helping generate **100,000+ total views.**

Piano
Pianist *3 hrs/wk, 50 wks/yr | 8 years*

- Studied piano for **8 years**, completing the Bastien Piano Basics curriculum through Level 4 and performing in 3–4 recitals annually.
- Perform monthly at **three senior-living communities**, using music as an ongoing form of community service.

Greenhill Basketball
Boys' JV Player & Girls' Varsity Practice Player *7 hrs/wk, 20 wks/yr | Grades 9–12*

- Played two seasons of boys' JV basketball; moved up to varsity early sophomore year during football season.
- Served as a **girls' varsity practice player** junior and senior year, providing live-game repetitions and additional competition during practices.

Greenhill JV Golf
JV Golfer *10 hrs/wk, 10 wks/yr | Sophomore Year*

- Competed on Greenhill's JV golf team for one season while balancing basketball and other extracurricular commitments.

Additional Digital Media Projects
Independent creative work

- Created short-form anime video edits on TikTok, with posts receiving more than 30,000 combined views. @meesin_editss2, 3 hrs/wk, 15 wks/yr
- Help run a casual food-focused Instagram account with friends, taking responsibility for creating, editing, and publishing posts from restaurants and outings. @bbufoodreviews, 1 hr/wk, 15 wks/yr

TECHNICAL & CREATIVE SKILLS

Programming: Python, Java, JavaScript, R
Machine Learning / Data: scikit-learn, PyTorch, Pandas, NumPy, computer vision
ECE / Research: FFTs, spectrograms, signal-processing simulation, Raspberry Pi, Linux/Ubuntu
Web / Software: React, Flask, MySQL, Git/GitHub
Design / Media: Adobe Premiere Pro, After Effects, Figma